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Title: "Analysis of Breast Cancer Wisconsin (Diagnostic) Data Using Logistic Regression Modeling"
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Introduction

Our study investigates whether logistic regression can predict breast cancer effectively. We explore two main questions: Can logistic regression accurately forecast breast cancer across diverse patient features? Do factors like tumor type and patient demographics enhance prediction accuracy? Our goal is to validate logistic regression's reliability for diagnosing breast cancer, which is crucial for successful treatment. We'll examine how well this statistical model predicts the disease by analyzing a range of patient data. This work could improve diagnosis and patient care.

Data Description

Our study's data comes from Kaggle and features breast cancer cases. The key elements include:

- Source: Kaggle.com.
- Response Variable: 'Diagnosis', coded as 0 (benign) or 1 (malignant).
- Predictors: Tumor characteristics like radius, texture, area, perimeter, and smoothness.
- Dimensions: 569 rows (cases) by 32 columns (variables).

Summary Statistics and Graphical Displays:

- Mean Tumor Radius: 14.13 units (SD: 3.52 units).
- Average Texture: 19.29 units.
- Average Perimeter: 91.97 units.

Graphical Displays:

Figure 1: Distribution of Diagnosis

The histogram illustrates a significant imbalance in the dataset, with a higher occurrence of benign cases compared to malignant ones. This indicates a predominance of benign diagnoses within the data.

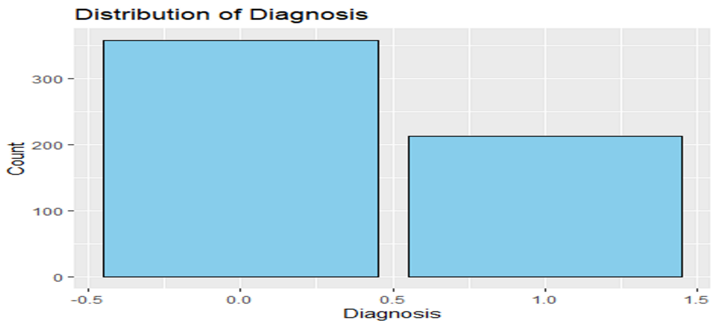
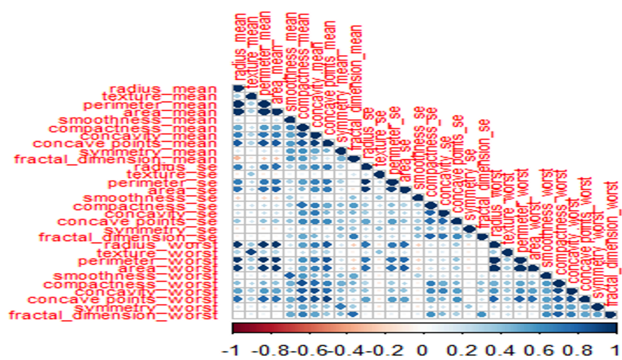


Figure 2: Scatterplot showing relationship among numeric variables

The correlation matrix provides valuable information regarding the interrelations between variables, facilitating the detection of multicollinearity. This insight is crucial for feature selection in modeling, ensuring the inclusion of independent and non-redundant predictors, thereby enhancing the model's predictive accuracy and interpretability.



Methods and Results

Feature Selection

We implemented a correlation filter to select features, excluding any with correlations above 0.7 to avoid multicollinearity, and prepared a refined set for modeling.

Model Training

We split our data into features (70%) and target (30%) sets, then trained a logistic regression model due to its interpretability and efficacy in binary classification.

Logistic Regression Model Summary

The model highlighted significant features like area_mean and various fractal dimensions, tying them to the likelihood of malignancy.

Figure 3: Coefficients Summary

```
Coefficients:
                         Estimate Std. Error z value Pr(>|z|)    
(Intercept)            -4.072e+01  6.174e+00  -6.595 4.25e-11 ***
texture_mean            2.673e-01  9.059e-02   2.950  0.00317 ** 
area_mean               2.059e-02  3.104e-03   6.634 3.28e-11 ***
symmetry_mean           1.342e+01  1.954e+01   0.687  0.49227    
texture_se              1.030e+00  8.088e-01   1.273  0.20297    
smoothness_se           2.513e+02  2.081e+02   1.207  0.22730    
symmetry_se             2.184e+01  6.020e+01   0.363  0.71680    
fractal_dimension_se   -1.016e+03  4.948e+02  -2.054  0.03994 *  
smoothness_worst        4.633e+01  2.909e+01   1.593  0.11125    
symmetry_worst          7.202e+00  9.840e+00   0.732  0.46420    
fractal_dimension_worst 1.336e+02  5.860e+01   2.280  0.02260 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

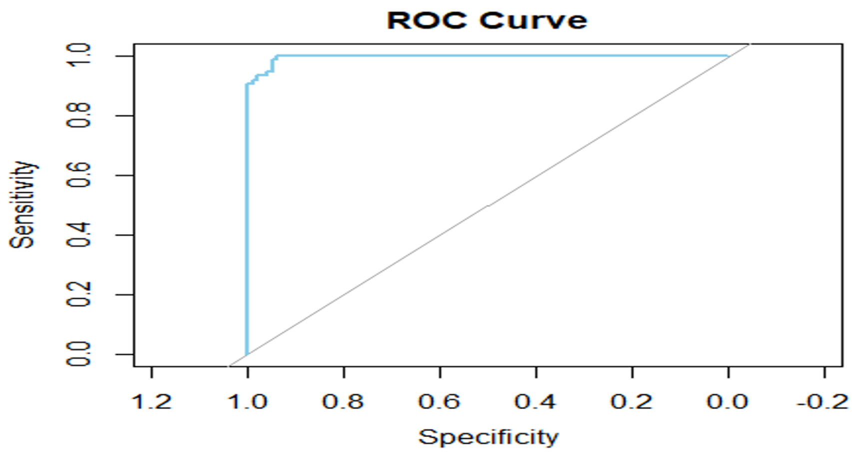
Model Evaluation

Metrics like accuracy (95.88%), precision (93.15%), recall (97.14%), and AUC-ROC (0.996) underscored the model's predictive strength.

Table 1: Evaluation Metrics

Metric	Value
Accuracy	0.9588235
Precision	0.9315068
Recall	0.9714286
AUC-ROC	0.996

Figure 4: ROC Curve



Confusion Matrix Interpretation

The matrix confirmed high true positive (95) and true negative (68) rates, with minimal false positives (5) and negatives (2), validating the model's reliability.

Table 2: Confusion Matrix

	Actual Benign	**Actual Malignant**
Predicted Benign	95	5
Predicted Malignant	2	68

Conclusion

Our regression analysis confirmed that logistic regression is a robust model for breast cancer diagnosis, effectively identifying key patterns and associations in patient data. The model's predictive accuracy was supported by significant performance metrics:

- Accuracy: 95.88%
- Precision: 93.15%
- Recall: 97.14%
- AUC-ROC: 0.996

These results suggest the model's potential utility in clinical settings, particularly in aiding biopsy analysis for tumor classification. However, further research is necessary to enhance the model's clinical applicability and to optimize patient outcomes in breast cancer care.

Code Appendix

The appendix would contain the R code used for the analysis.

```
# Visualize the distribution of selected features
selected_features <- c("radius_mean", "texture_mean", "perimeter_mean", "area_mean")
for (feature in selected_features) {
  ggplot(data, aes(x = !!sym(feature))) +
    geom_histogram(fill = "skyblue", color = "black") +
    labs(title = paste("Distribution of", feature),
         x = feature,
         y = "Frequency")
}

# Correlation matrix
correlation_matrix <- cor(data[, -c(1, 2)])  # Exclude ID and diagnosis columns
corrplot::corrplot(correlation_matrix, method = "circle", type = "lower", tl.cex = 0.7)
```

```
X <- data[, selected_features]  # Use only selected features
y <- data$diagnosis

# Split the data into training and testing sets (70% training, 30% testing)
set.seed(123)  # For reproducibility
train_index <- createDataPartition(y, p = 0.7, list = FALSE)
X_train <- X[train_index, ]
X_test <- X[-train_index, ]
y_train <- y[train_index]
y_test <- y[-train_index]

# Combine the target variable and selected features into one data frame
train_data <- cbind(y = y_train, X_train)

# Train a logistic regression model using the training data
logit_model <- glm(y ~ ., data = train_data, family = binomial(link = "logit"))
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
# Print the summary of the logistic regression model
summary(logit_model)
```

```
##5. Model Evaluation

# Predict on the testing data
predictions <- predict(logit_model, newdata = X_test, type = "response")

# Convert probabilities to class predictions (0 or 1)
predicted_classes <- ifelse(predictions > 0.5, 1, 0)

# Confusion matrix
conf_matrix <- table(predicted = predicted_classes, actual = y_test)
conf_matrix
```

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